

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1894

Blackrock 138 kV

May 15, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1894. The Interconnection Customer proposes a 155.8 MW battery energy storage facility interconnecting at the Blackrock 138 kV substation in the Pennsylvania area. The request was received on October 04, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 531 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,191,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Ciales 8701 to Sjve Madera 138 kV
Contingency: Blackrock to Shawville Station 345 kV
Loading: 112.4% (Rating: 187 MVA)
DFAX: 13.0%
Pre-project loading: 101.6%
2. Facility: Phoenix to Sjve Madera 138 kV
Contingency: Blackrock to Shawville Sta 345 kV
Loading: 108.0% (Rating: 202 MVA)
DFAX: 13.0%
Pre-project loading: 98.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ciales 8701 to Sjve Madera 138 kV	Rebuild line with higher-capacity conductor	\$18,752,000	72% (\$13,424,000)
Phoenix to Sjve Madera 138 kV	Reconductor line	\$2,468,000	72% (\$1,767,000)
	Total allocated to project		\$15,191,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.